**Reviving Online Sales: Addressing Order Decline in a Biryani Cloud Kitchen**

**Final Report**

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## 1. Executive Summary

This project is carried out in collaboration with **Ax-Wxxx Bxxxxxx**, a B2C cloud kitchen based in **Othakadai, Madurai**. Despite offering competitive pricing in the biryani segment, the business has faced **stagnant growth and low brand rating since mid-2023**. Key challenges are **lack of customer retention strategy**, limited understanding of customer preferences and competitors, **delivery issues during peak hours**, and Zomato's algorithm challenge occasionally applying discounts without consent or marking the restaurant offline because of driver unavailability.

To address these concerns, **sales data** from the Zomato merchant dashboard, **customer data** inferred from encoded IDs, and **competitor data** scraped from Google Maps and Zomato were analyzed. The cleaned datasets underwent Exploratory Data Analysis (EDA), descriptive statistics, and time-series modeling using SARIMA to capture weekly sales seasonality. Customer behavior was assessed through churn rate, bill value distribution, and sentiment analysis.

The analysis revealed **higher weekend order values** despite larger weekday volumes and a **high customer churn rate of 96%**. Competitor analysis indicated significant overlap in menu and pricing, contradicting the owner's belief of market uniqueness in Chennai-style biryani. These insights guided actionable recommendations on **customer retention, promotion strategy, and brand positioning**. Implementation of these findings is expected to enhance **sales consistency, customer loyalty, and overall brand image**.

## 2. Detailed Explanation of Analysis Process/Method-

### 2.1 Data Cleaning and Preprocessing

The first stage of the analysis involved data cleaning and preprocessing steps to ensure that all datasets were accurate, consistent, and suitable for in-depth analysis. The Zomato sales dataset underwent several refinement steps, beginning with the updating datatype, removal of missing values, ambiguous symbols, and redundant identifiers such as restaurant name, ID, subzone, and city. This ensured that subsequent analysis focused only on variables directly linked to order performance and customer trends, aligning with the goal of understanding sales stagnation and operational inefficiencies.

Additional time-based features such as weekdays, weekend,day, month, year, quarter, and half-year were extracted from the "Order Placed At" column, while the time component was excluded since the restaurant operated only during noon hours. This preprocessing step was essential for uncovering **fluctuations in demand across different periods** and for identifying potential growth-related issues in order patterns over time. A customer classification feature was also introduced to separate repeat and non-repeat buyers by tracking unique order counts per customer ID. This is to study **poor customer retention**, a key issue faced by the business.

To simplify promotional impact analysis, multiple discount-related fields were consolidated into a single variable representing whether an order was placed with or without a discount. This ensured consistency in evaluating how Zomato's automatic discounting was influencing purchasing patterns. Likewise, grouping orders into defined price ranges (below ₹200, between ₹200–₹299, and ₹300 or above) provided a structured way to study how voucher distribution affected sales across spending levels, addressing the question of unaccounted or mispriced discounts.

Since structured competitor datasets were unavailable, a **custom web scraping method** was used to collect data from Zomato and Google Maps. Each listing's name, rating, cuisine, price, and delivery time was extracted using **BeautifulSoup**, chosen for its efficiency in parsing semi-structured HTML data. This approach allowed the collection of competitor insights essential for benchmarking Ax-Wxxx's pricing and performance against other competitors.

The initial dataset of 246 listings was refined using filters to isolate true competitors i.e., non-vegetarian restaurants offering biryani (excluding Seeraga Samba and Hyderabadi variants) priced below ₹400 with delivery times under 40 minutes. This process narrowed the list to 41 competitors operating within Ax-Wxxx's target market and service area. A custom Google Map is created to visualize their geographic distribution, revealing competition hotspots, delivery feasibility, and opportunities for improvements in delivery service.

For customer reviews, data was manually extracted using **Windows PowerToys Text Extractor**, as Zomato does not permit direct review downloads. The extracted text was cleaned by removing emojis, special characters, and stopwords using **NLTK**, ensuring linguistic uniformity for later sentiment analysis. Reviews were then categorized as positive(>=4), neutral(3), or negative(<=2) based on ratings, providing a measurable view of customer satisfaction patterns that could explain the brand's low visibility and retention issues.

Further orders contained multiple items per transaction, so unnormalized the "items in order" field and categorized items into logical groups (Biryani, Combo,Bucket, Starters, Drinks) using keyword-based classification. This categorization made it possible to connect sentiment trends to specific menu types and identify which offerings drove positive feedback or repeat orders.

Overall, each step of data cleaning and preprocessing is strategically aligned with understanding declining engagement, identifying reasons for low customer retention, and benchmarking against competitors.

**2.2 Analysis Process/Method**

To comprehensively understand the nuances of Ax-Wxxx's business, extensive data exploration and analysis (EDA) were conducted to study customer behavior & sentiment, impact of discounts and offers, historical sales trends, and menu performance. The analytical framework integrated time series forecasting, statistical transformations, natural language processing, network-based co-occurrence analysis, and correlation heatmaps. Each method was carefully selected based on its relevance to the dataset and the specific insights required for strategic decision-making.

#### *2.2.1 Time Series Analysis*

To understand the sales trend and recurring order patterns over time, a **Seasonal ARIMA (SARIMA)** model was used. The decision to use SARIMA was made after observing **7-day, 14-day seasonal cycles** in autocorrelation plots, especially noticeable peaks during weekends which could not be captured by simpler linear models.

Based on this, the SARIMA model was configured to include a **weekly seasonal component**, enabling it to forecast future order volumes more accurately while accounting for both trend and seasonality. The forecasts generated from this model can guide **staff scheduling, kitchen preparation planning, and delivery resource allocation**, particularly during weekends when both demand and driver shortages are expected to peak.

In addition, **Kitchen Preparation Time (KPT)** was analyzed to verify the owner's claimed benchmark of around 3 minutes per order (as food was mostly pre-cooked). Outliers were found extending up to 20 minutes, which pointed to possible inefficiencies or order-handling delays. Tracking this metric over the last six months revealed fluctuations in preparation consistency, providing useful evidence for **employee training and process improvement**.

#### *2.2.2 Distribution Analysis and Transformation*

Many numerical variables (bill amount, KPT) were right-skewed, which can bias models and inference. To stabilize variance and improve model assumptions applied a log transform:

$$\text{log\_bill} = \log(\text{bill\_value} + 1)$$

This transformation reduces the influence of extreme bills and makes comparisons across segments (weekend vs. weekday; discount vs. no discount) more robust. The choice is justified because of the need for reliable statistical tests and forecasting unaffected by heavy tails when measuring the real effects on revenue.

#### *2.2.3 Menu Analysis and Co-Occurrence Network*

To discover natural bundling of menu items and cross-sell opportunities, built a co-occurrence network where nodes = items and weighted edges = co-order strength. Instead of relying on raw co-occurrence counts alone, created a matrix to map correlations between items i.e.,"Does ordering item A increase the likelihood of ordering item B?".This clarified which dishes are naturally ordered in pairs or groups, guiding better menu bundling and promotional decisions to improve average order value (AOV).

Also, the restaurant's Zomato listing (images and descriptions) was reviewed to spot any interface or content gaps that might be affecting conversions. It's a quick but impactful step to strengthen order rates.

#### *2.2.4 Sentiment Analysis*

For customer reviews, basic NLP preprocessing steps such as lowercasing, emoji and special character removal, and stopword filtering using NLTK was carried out due to its reliability with small-to-medium datasets and transparency in processing steps.

To further visualize relationships between **quantity, distance, and ratings** used heatmaps to identify whether delivery delays or longer distances consistently led to negative feedback. Item-level sentiment using bar graphs further revealed dishes with no positive ratings directly indicating for recipe or process improvement, or even de-listing if required.

#### *2.2.5 Customer Retention and Segmentation*

Customer IDs were preprocessed and grouped into repeat and non-repeat categories to compare their ratings and average order value (AOV). A cohort-based model was then applied to study customer retention and lifetime value, where each customer was assigned a cohort based on their first purchase month. A cohort matrix was constructed to track retention patterns and derive key metrics such as Customer Churn Rate (CCR) and Repeat Purchase Rate (RPR).

Further, Customer Lifetime Value (CLV) was computed by aggregating each customer's total bill amount over their active period and normalizing it by the number of active months: CLV = (Total Bill Value over Lifetime) / (Active Months).

This measure helped identify **high-value customers** suitable for loyalty and **retention programs.** Trends over the last six months were analyzed through frequency distributions to understand variations in ordering behavior across different customer segments.

#### *2.2.6 Competitor Analysis*

Competitors were categorized by **Biryani type** such as Basmati (BS, cluster of Ax-Wxxx), Hyderabadi(HYD), Seeraga Samba(SS), and others to understand broader market landscape and positioning. **Pricing distribution** and **restaurant ratings** were compared to benchmark performance and identify differentiation gaps. Also for plotting Power BI was used for ease of sharing with business owners.

A **geographic mapping** of competitors was carried out to assess the advantage of proximity and hotspot clustering, as most orders originate within a **5 km radius**. This spatial view also supported understanding the effect of **driver availability** and delivery zone concentration on order completion rates.

## 3. Results and Findings

### 3.1 Time Series and Temporal Patterns

*A. Yearly Sales Comparison*

At the yearly level, total net bill value in 2024 showed a **98% growth** over 2023, indicating a strong expansion in both **customer base** and **order frequency**. This reflects successful market capture and increased transactions over time.

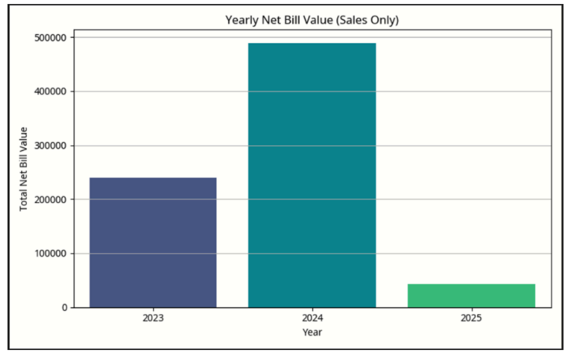



*Figure 1 – Yearly Net Bill Value (Sales Only)*

Despite the near-doubling in annual sales, the **business was perceived as stagnant**, largely due to **inflationary pressures**, **rising raw material costs**, and **increased delivery and marketing expenses**, which diluted visible profit margins. Additionally, **staff salary adjustments** and **higher third-party delivery commissions** may have further offset perceived financial gains. As a result, while top-line revenue displayed substantial momentum, **bottom-line pressures** contributed to a misleading impression of flat performance.

The 2025 data was excluded from this comparison, as only January transactions were available, which would not provide a representative view of the full year's performance.

*B. Monthly Sales Trend Analysis*

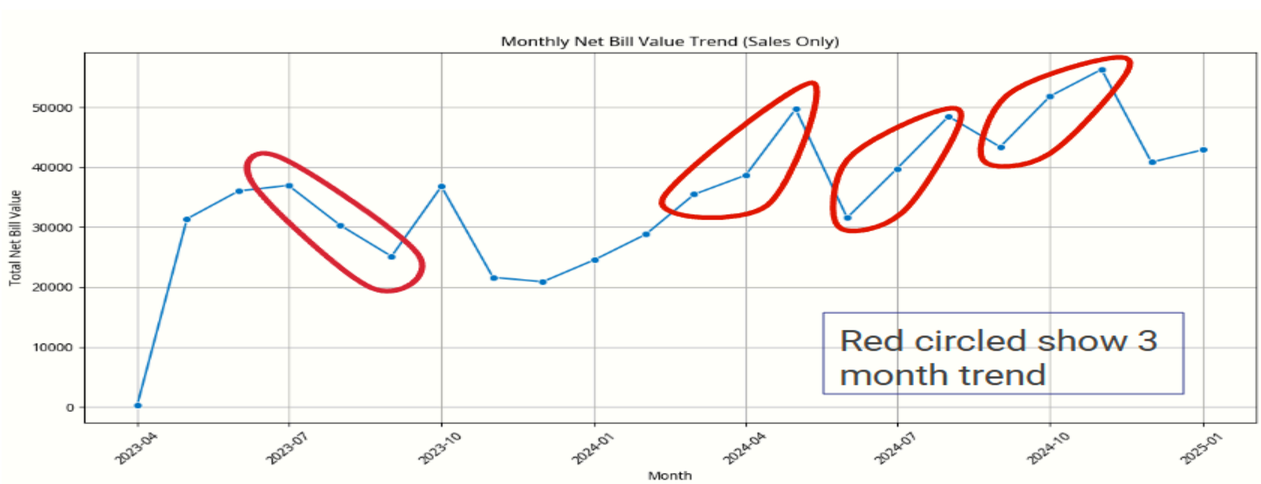



*Fig 2 - Monthly Net Bill Value Trend Chart*

The monthly trend in *Fig. 2* reinforces the annual findings, showing a steady upward trajectory in total net bill value from the end of 2023 through 2024. Following the initial launch phase, sales began stabilizing from April 2023 and displayed consistent growth throughout 2024 except

September and December showing dips in both years, marking an estimated **70–80% increase** from the previous year's monthly averages.

A closer observation reveals a **three-month recurring pattern**, where sales typically rise for two consecutive months before dipping in the third. This cyclical pattern likely reflects **Holiday seasons** and **festival-linked demand surges**. While these dips may have momentarily reinforced the

perception of stagnation, the broader trend indicates **sustained growth** over time. Understanding this periodic rhythm provides actionable insights for business planning particularly to **align marketing campaigns, loyalty offers, or limited-time promotions** during low-demand periods to maintain steady order volumes.

Complementing the trend analysis, **monthly box plots** (Fig. 3) showed consistent median bill values across the year, with only a few high-value outliers in **May, July, and November** likely linked to festive or holiday peaks. This consistency suggests a **stable customer spending pattern**, supporting operational predictability and inventory planning.

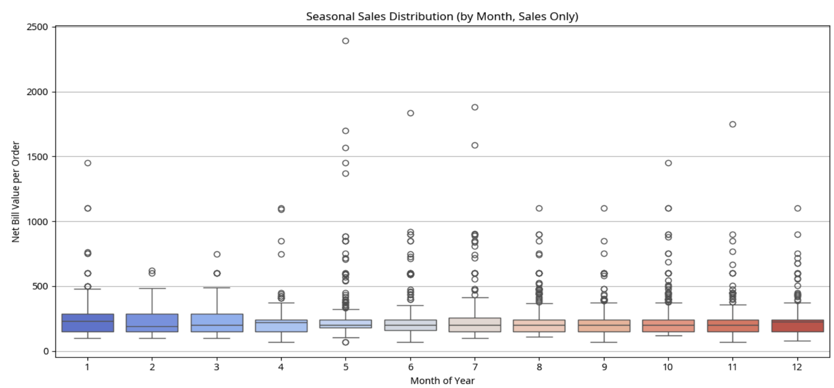


*Fig 3 – Box Plot of Total Sales amt Distribution by Month*

*C. Short-Term Demand Forecasting*

To identify temporal trends and predict short-term demand, a Seasonal ARIMA (SARIMA) model was employed using daily sales data. Autocorrelation (ACF) and partial autocorrelation (PACF) plots revealed recurring peaks at lags of 7 and 14 days, indicating strong weekly cyclicality in order patterns. Based on these observations, a SARIMA(1, 0, 1)(1, 1, 1, 7) model configuration was adopted, incorporating a seasonal component of seven days to reflect this rhythm in customer behavior.

These findings reinforced that order volumes on the Zomato platform follow a structured weekly rhythm rather than purely random fluctuations, emphasizing the importance of aligning kitchen operations, staff work schedules, and delivery capacity with predictable demand cycles.

*C.1 Model Fit and Validation Summary*

Model validation on the test dataset yielded a Mean Absolute Error (MAE) of **594.94** and a Root Mean Squared Error (RMSE) of **748.32**, suggesting that daily forecasts deviated by approximately ₹600–₹750 on average. These deviations are acceptable given the natural volatility in online food delivery, where order counts can vary widely due to promotions, weather, or holidays.

The Mean Absolute Percentage Error (MAPE) appeared disproportionately high, primarily because of several zero or near-zero sales days in the test period. Since MAPE is sensitive to small denominators, it tends to exaggerate error in such contexts and is therefore not a reliable indicator of model performance.

Despite these limitations, the model effectively reproduced the overall weekly sales pattern. The residual variations were random rather than systematic, confirming that the SARIMA model provided a sound representation of seasonality. Hence, it serves as a practical forecasting tool for planning kitchen production, raw material procurement, and workforce allocation, especially during anticipated high-demand weekends.

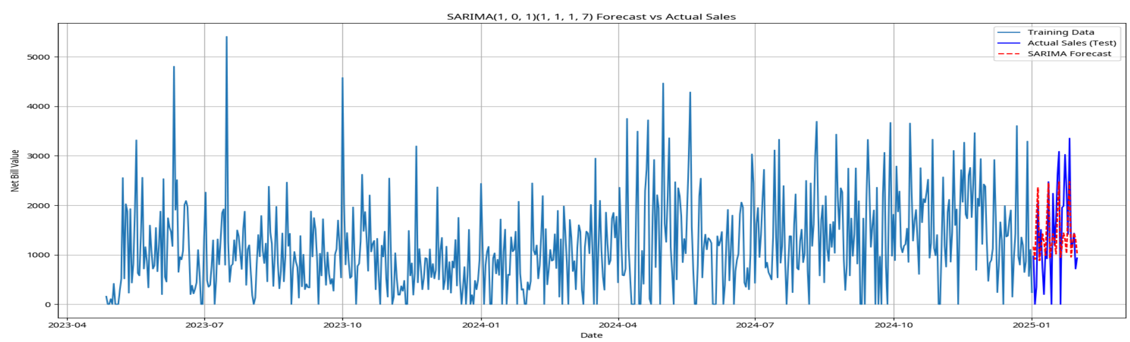



*Fig 4 - Time Series Plot [SARIMA Forecast]*

### *D. Weekday vs Weekend Performance Analysis*

Building on the observed seven-day seasonality, a comparative analysis was performed between **weekday and weekend** performance to uncover behavioral distinctions in customer ordering.

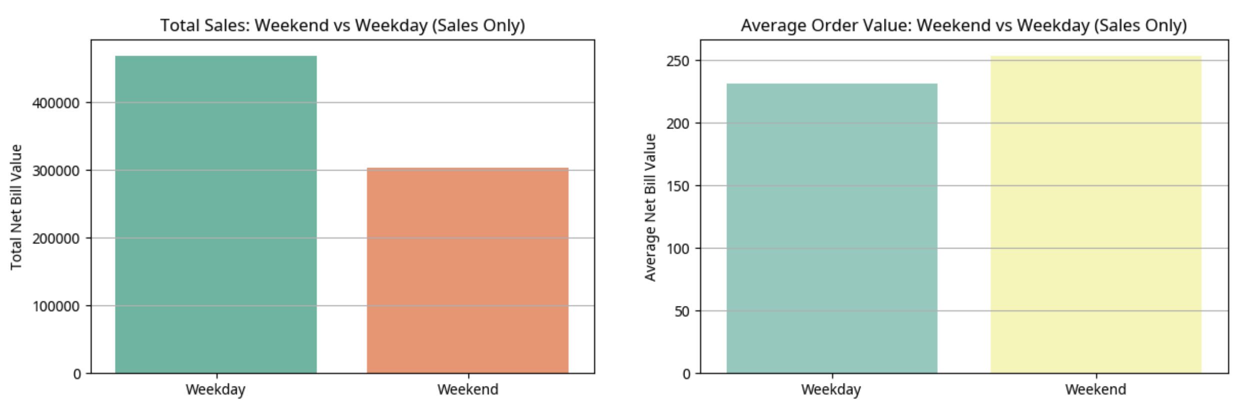



*Fig 5 - Weekday Vs Weekend Plot [Left graph:Total Sales, Right graph:AOV]*

The findings indicate a **clear divergence between order volume and average bill value** revealing two distinct consumption patterns:

- **Weekday Routine Consumption:** Higher total sales driven by numerous low-to-medium-value individual orders, typically single-meal or lunch purchases from regular customers within a 5 km radius.

- **Weekend Indulgence Behavior:** Fewer but higher-value orders, often family or group meals featuring combos or biryani buckets, resulting in greater average bill values despite lower transaction volume.

From a strategic perspective, this distinction highlights an opportunity to fine-tune promotional efforts where **weekday promotions** should focus **on driving order volume** (e.g., quick-meal combos or lunchtime offers and **Weekend campaigns** should emphasize **value bundling** and **premium experiences**, targeting higher order sizes and family-oriented orders. extending to day wise, revealed Sunday and Wednesday to be popular days. These insights can support data-driven planning of discount schedules, staffing allocation, and targeted marketing interventions to smooth demand and maximize operational efficiency.

### 3.2 Menu Analysis and Customer Preferences

The Menu analysis aimed to determine which menu items drive repeat sales, which combinations occur most frequently, and where customer confusion or unmet expectations might exist.

#### *A. Item Performance Overview*

Sales concentration was found to be heavily skewed toward a few key dishes, particularly the **Biryani category**, which accounted for the highest volume and consistent weekly demand. Low-performing SKUs such as single-piece biryanis and Double egg biryani variants showed minimal contribution without substantial revenue impact. Streamlining the menu could reduce preparation load and ensure consistent quality during high-traffic periods identified in the SARIMA analysis.

A detailed breakdown of the Biryani category confirmed strong customer preference toward familiar, value-driven variants such as Chicken Biryani [Regular] & Chicken Biryani [1 kg]. However, items like Special 65 Chicken Biryani (₹199) and 65 Chicken Biryani (₹189) showed

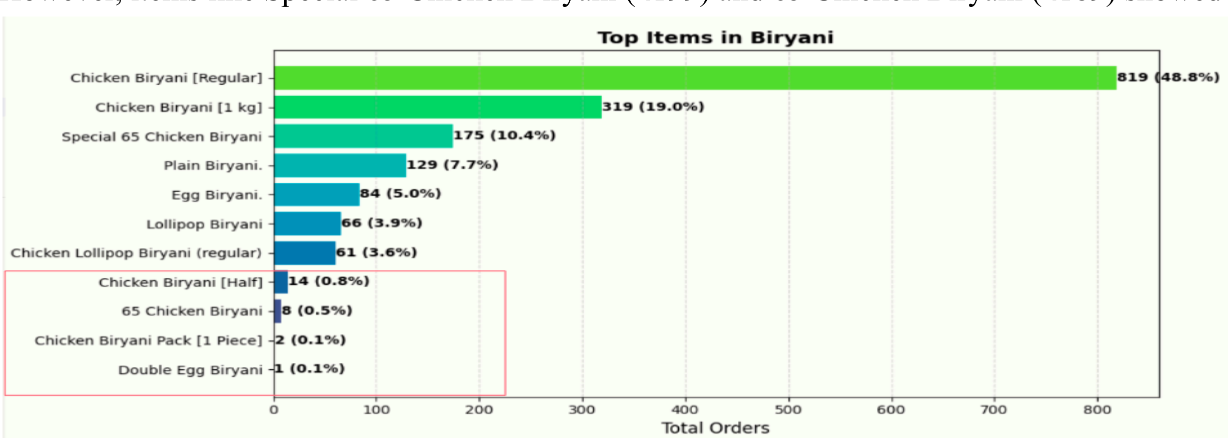



*Fig 6 - Ranking of Menu Items under Biryani Category by Order Volume* demand fragmentation despite minimal price differentiation (Total 175 and 8 orders respectively). These closely priced variants appear to cannibalize each other without enhancing overall sales, indicating potential for rationalization. Similarly, Chicken Biryani Pack (1 Piece), Double Egg Biryani, and Chicken Biryani (Half) displayed negligible demand and limited positive sentiment, adding operational complexity without meaningful contribution to revenue. **Streamlining these low-performing SKUs** can simplify kitchen workflow and sharpen focus on high performers.

*B. Portion-Size Ambiguity and Its Impact on Ratings*

Customer sentiment data provided additional insight into perceived value and satisfaction. Fig 7 (Sentiment Distribution for Starters) shows that items under the *Chicken 65* category received mixed feedback, with sentiment polarity often linked to portion clarity (from Section 3.3 C). The lack of distinct differentiation between *Chicken 65*, *Chicken 65 [120g]*, *Chicken 65 [250g]*, and *Chicken 65 [500g]* appears to have created confusion. When **identical product images and vague descriptions** are used across different portions, customers tend to select the lower-priced option, expecting equivalent quantity. This mismatch between perceived and actual portion sizes likely contributed to inconsistent ratings.

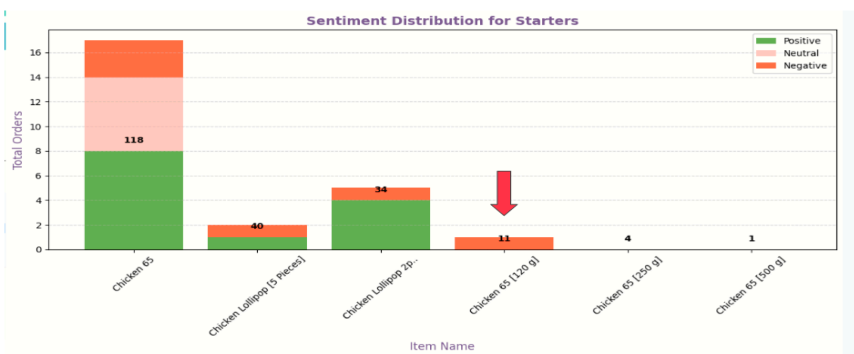



*Fig 7 - Sentiment Distribution of Starters [Left]; In-App Menu Listing Image [Right]*

Emphasizing portion clarity through **detailed menu descriptions and unique product visuals** can enhance transparency and increase average order value through informed upselling.

*C. Co-occurrence and Cross-Item Relationships:*

The co-occurrence heatmap (Fig 8) revealed strong pairing tendencies between Regular Chicken Biryani and Chicken 65. Interestingly, despite these items already being available together in combo offerings, customers frequently ordered them separately. This behavior suggests a **psychological perception that combo portions are smaller or less satisfying or not value for money**. Introducing portion transparency such as indicating “3-piece Chicken 65 Combo” or “serves 1” along with weight-based labels may reduce confusion and improve combo uptake.

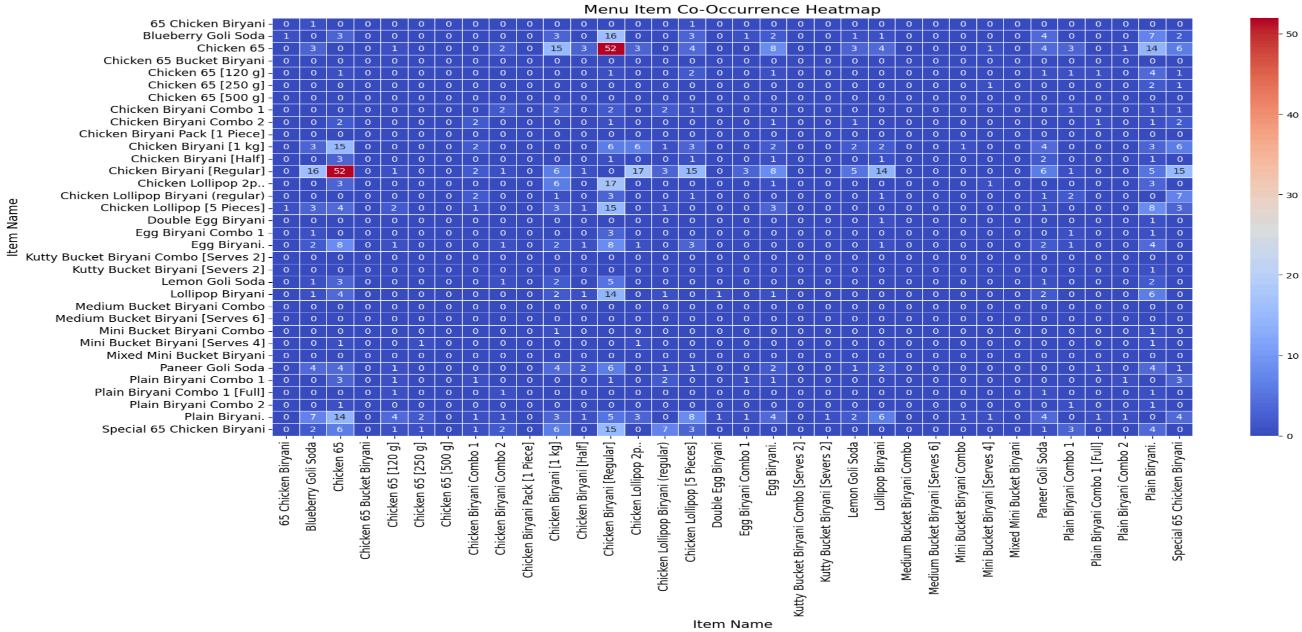



*Fig 8 - Co-occurrence Heatmap [Menu Items]*

At the same time, **correlation analysis** indicated weak or no positive relationship between *Regular Chicken Biryani* and *Chicken 65*, implying that **ordering one does not predict ordering the other**. This disconnect could be driven by budget-conscious behavior among single-item buyers. Introducing low-risk trial incentives such as a complimentary two-piece *Chicken 65* sample for first-time customers could improve satisfaction and increase future combo adoption.

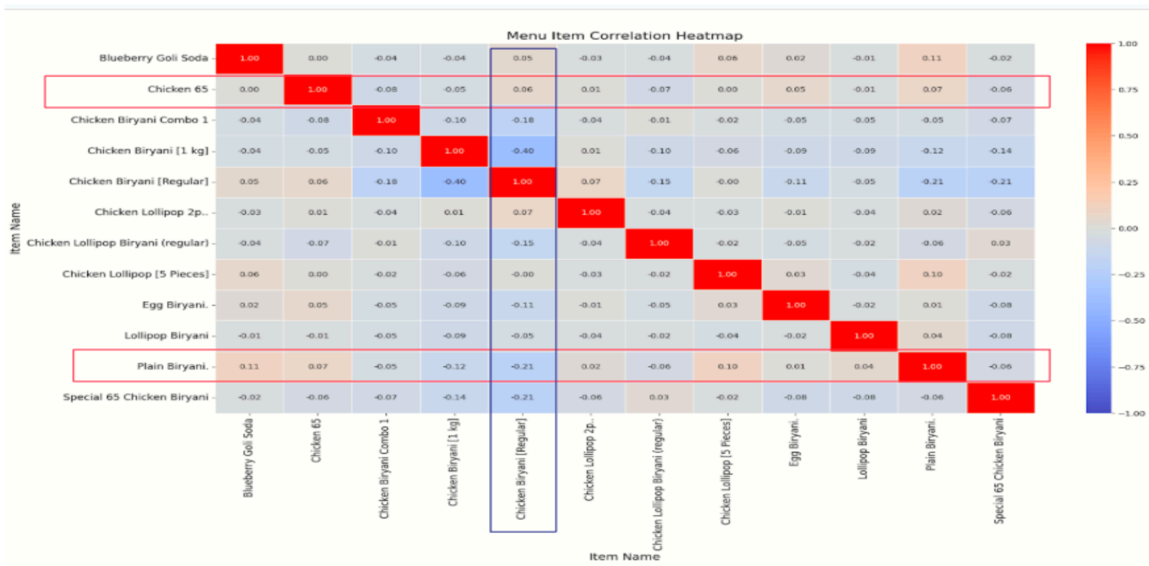



*Fig 9 - Correlation Heatmap [Menu Items]*

*D. Cross-Category Patterns and Seasonal Behavior:*

An interesting behavioral pattern was found in the Chicken 65 network diagram (Figure 10). Customers who ordered **Chicken 65 Biryani often added an additional serving of Chicken 65**, signaling strong loyalty to that specific flavor profile. This opens the possibility for a data-informed menu innovation, such as a “Double Chicken 65 Special”, that aligns with demonstrated customer preferences.

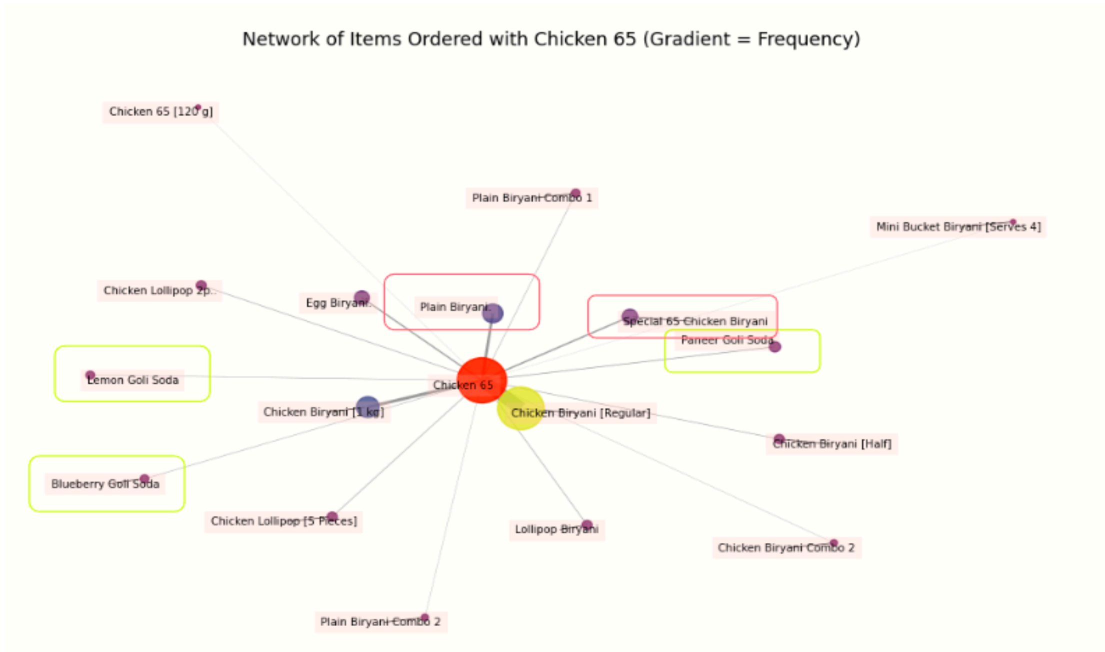



*Fig 10 - Network Graph of items ordered with Chicken 65*

Seasonal variations were also evident. **When Chicken 65 was ordered, the demand for soft drinks dropped considerably**, likely due to the winter season’s reduced beverage consumption. This trend warrants stock monitoring during the summer months to validate whether the decline is seasonal or product-specific. If seasonality holds, the beverage menu can be changed seasonally with warm alternatives like soups during colder months to maintain add-on sales volume.

*E. Drinks and Combo Category Insights:*

Although *Paneer Goli Soda* recorded lower sales than *Blueberry Goli Soda*, it consistently maintained ratings above 4, whereas *Blueberry Goli Soda* received lower average ratings, often below 3. The two beverages belong to different brands, which makes it essential to **monitor brand reputation** periodically to ensure consistency in quality and perception. While no explicit taste-related complaints were recorded, the lower satisfaction associated with *Blueberry Goli Soda* could stem from a combination of factors like **brand-related reputation issues, subtle aftertastes that may not pair well with biryani**, or the **novelty-seeking behavior** of customers. Many might have initially chosen *Blueberry Goli Soda* out of curiosity, preferring something different from the familiar Soda flavor, but later found it incompatible with the flavor profile of biryani, contributing to its lower repeat preference and ratings.

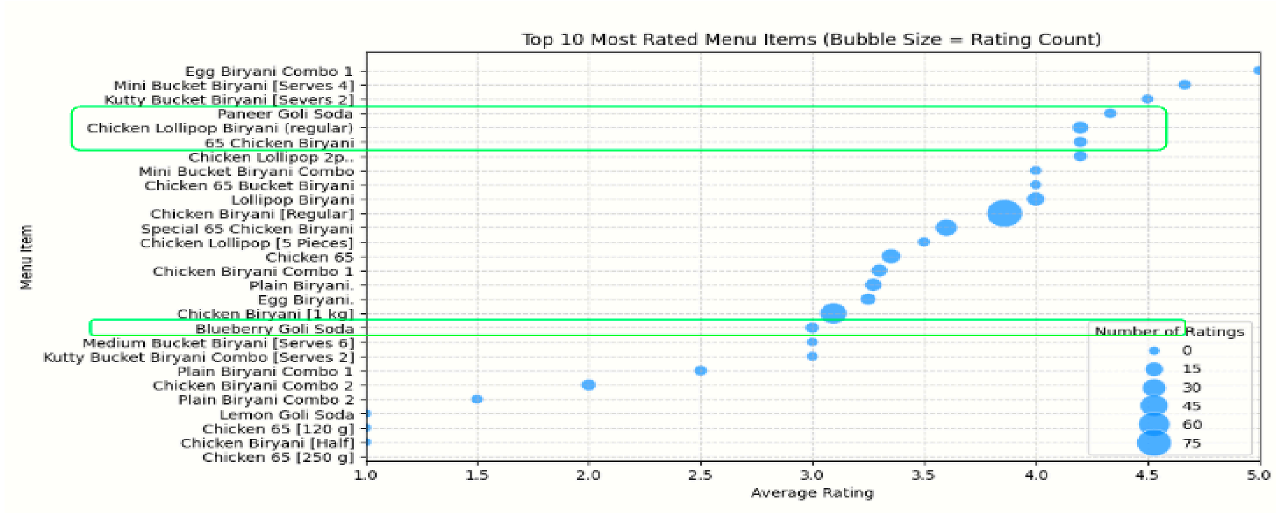



*Fig 11 - Scatter plot of Menu Vs their Average Rating with Bubble indicating the # of ratings*

Conducting **occasional taste tests** and **cross-pairing trials** can help determine whether the aftertaste or brand differences influence customer satisfaction. To improve customer experience and optimize sales performance, following points are suggested:

- Promote both sodas as **add-on recommendations** in app alongside relevant food pairings using a *"Tastes Better With"* tag for better contextual appeal.
- Highlight *Paneer Goli Soda*'s strong customer ratings in promotional visuals to reinforce positive sentiment and drive choice confidence.
- Periodically monitor *Blueberry Goli Soda's* **brand reputation and sensory alignment** through in-store feedback and controlled taste checks to ensure product experience remains consistent with customer expectations.

*F. Biryani & Combo Menu Insights*

None of the popular items(Top 6 items in Fig 12) have achieved a 4+ rating, indicating room for improvement in recipes or presentation. Lollipop shows strong potential in biryani orders but has a noticeable rating gap between the 2-piece and 5-piece portions. Ensuring portion consistency, balanced pricing, and aligned customer expectations can improve satisfaction and ratings.

Although both 65 Biryani (₹189) and Lollipop Biryani (₹155) include the same accompaniments i.e., Brinjal curry, Raita, and similar portion size, **customers still preferred the Special 65 Biryani despite its higher price and comparatively lower rating**. One likely factor could be the item image: Special 65 Biryani is shown with curry, while Lollipop Biryani is not. Updating the Lollipop

Biryani image to display curry or clarifying accompaniments in its description may encourage more orders.

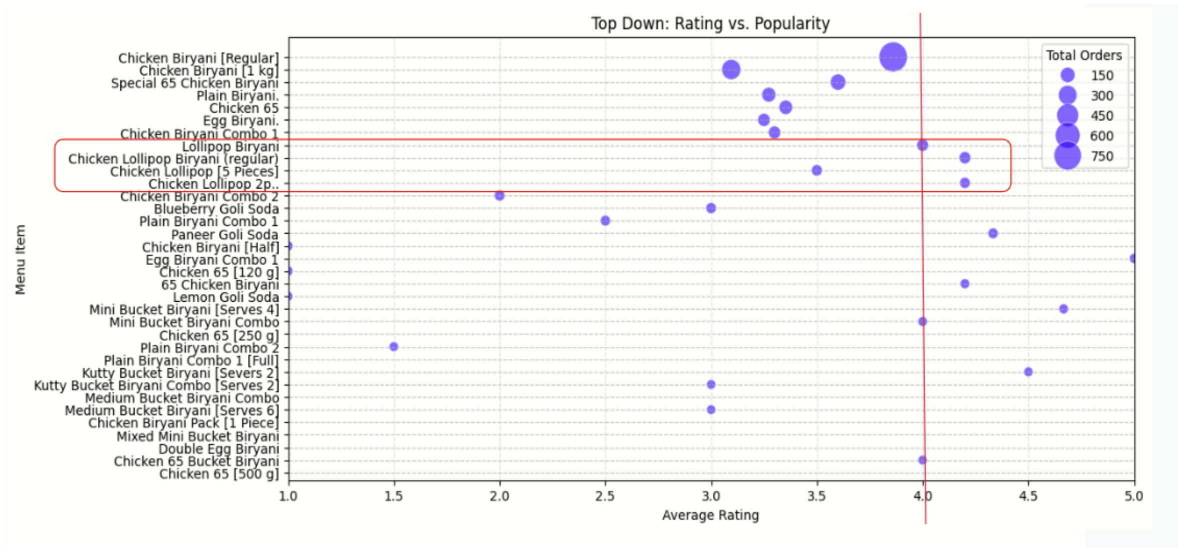



*Fig 12 - Scatter plot of Menu Vs their Average Rating with Bubble indicating the # of orders (Sorted)*

Interestingly, correlation mapping (Fig 9 in page 12) further revealed that **redefining combos solely based on logical food pairings (e.g., Regular Biryani with Lollipop)** would not substantially increase cross-item sales. Customer preferences seem guided more by **context, presentation, and trust** than by combination logic alone. This insight was reinforced by the finding that, except for Chicken Biryani Combo 1, most of the combo variants received a higher

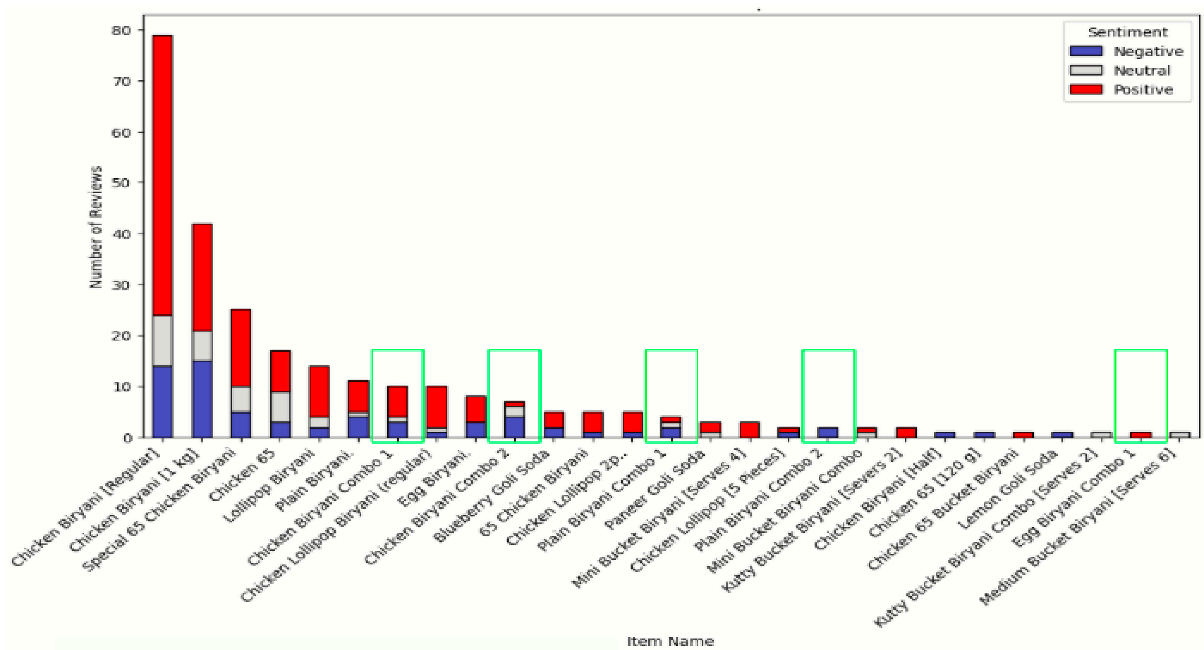
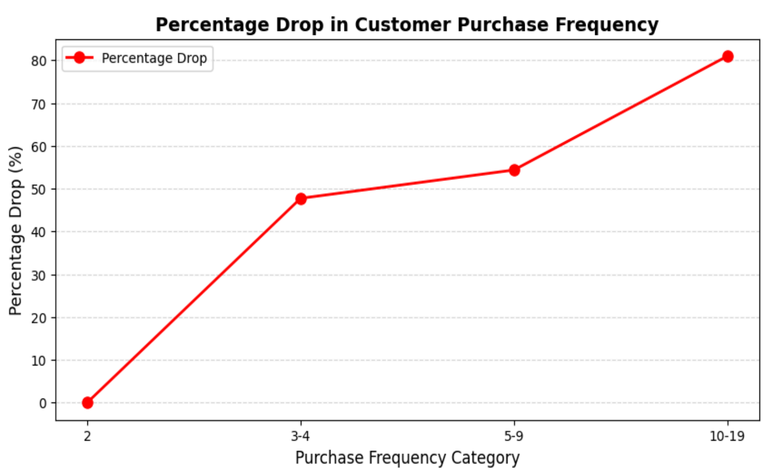



*Fig 13 - Items Vs Reviews with Sentiment* share of negative reviews despite including top-rated starter items. Such disconnects likely stem from **portion perception, presentation, and labeling clarity** rather than taste (Inferred from 3.3 Section C). Enhancing combo visuals, standardizing portion communication, and **distinguishing combo packs from standalone dishes** could help restore customer confidence.

### 3.3 Customer Behavior Analysis

Customer retention emerged as a key concern, with a high churn rate of 96.43% and very few customers making repeat purchases across multiple months. On the other hand, the average Customer Lifetime Value (CLV) currently stands at ₹317.20, revealing significant potential for growth through improved retention and engagement**.**

*A. Customer Retention Overview*




Retention remains low across all observed months, with very few customers making frequent purchases beyond their initial orders. Most disengage after one or two transactions, leading to a churn rate of **96.43%** and an average **Customer Lifetime Value (CLV)** of only **₹317.20**. The 49% drop after two orders highlights retention issues.

*Fig 14 Trend of % drop of Customer in each category (Top)*

The data also suggests that the business attracts new customers effectively but struggles to sustain long-term engagement. The limited number of **high-frequency buyers (>2 orders)** highlights a gap in loyalty-building mechanisms.

If customers discontinue after a few purchases, **pricing perceptions, delivery consistency, and competitive alternatives** may be key deterrents. Addressing these factors especially by **improving consistency, communication, and trust-building touchpoints** will be essential for boosting retention and increasing lifetime value.

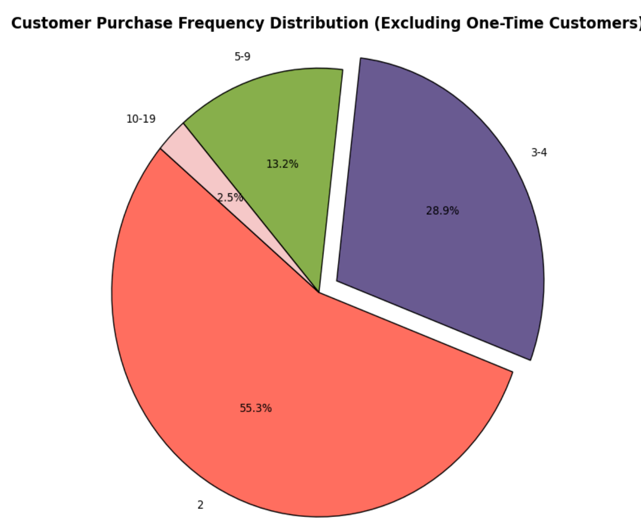



*Fig 15 (right) - Pie of Repeat Customer Purchase frequency with each pie showing order bin (eg: range 3-4 means from minimum 3 orders to maximum 4 orders)*

*B. Customer Type Comparison: Bill Value, Distance, and Ratings*

Customer segmentation reveals that 76.6% are one-time (non-repeat) customers, while only 23.4% are repeat customers. This imbalance underscores the importance of converting first-time buyers into repeat patrons to stabilize long-term revenue.

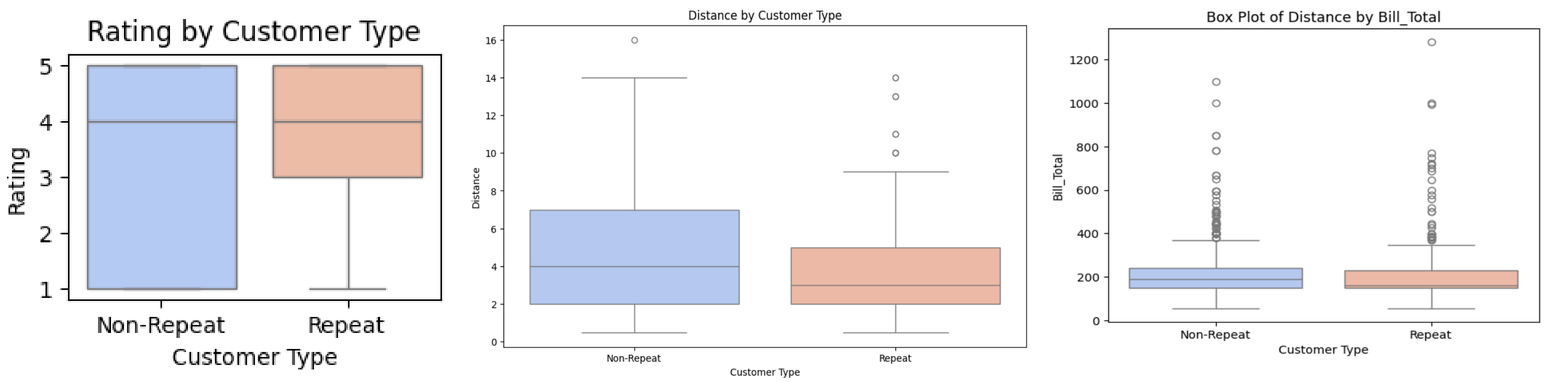



*Fig 16 Customer Type Vs Rating (Left),Customer Type Vs Distance (Middle), Customer Type Vs Bill Value (Right)*

Repeat customers consistently give higher ratings, reflecting positive and reliable experiences, while non-repeat customers show wider variation, might be due to inconsistencies in food quality, delivery experience, or expectations. Most repeat buyers order from within 2–4 km, highlighting the role of speed, convenience, and low delivery costs. In contrast, non-repeat customers often from beyond 4 km likely place trial orders influenced by low price but don't return due to delays or service quality. Although both groups spend similarly on average, non-repeat customers show more high-value outliers, indicating missed retention opportunities that could be addressed through targeted post-purchase engagement to boost Customer Lifetime Value (CLV) & growth.

*C. Customer Sentiment and Pain Points*


*Fig 17 Word Cloud from Customer reviews*

Sentiment analysis reveals two dominant feedback clusters like quantity-related concerns (word cluster- piece,small,less) and Good (positive satisfaction) highlighting clear contrasts in customer perception. Most negative mentions are linked to **portion size accuracy(40%), taste consistency(30%), and wrong or incomplete orders(10%).**

One possible reason for **wrong or incomplete orders** could stem from Ax-Wxxx's linear order workflow: the Order Taker notes down incoming orders manually on paper from App, after which the Packing Staff packs the food and hands over the order to the Delivery Partner for dispatch. This flow involves manual note taking and lacks an intermediate 3rd person verification step, increasing the risk of order mismatches or missing items.

Although sentiment clusters featured keywords like “tasty,” “good,” and “delicious,” the proportion of negative experiences indicates a need for operational refinement.

### 3.4 Competitor Analysis

The business was originally positioned around offering **Chennai-style Basmati Biryani (BS)**, under the belief that the local market in Madurai was dominated by **Seeraga Samba(SS)** and **Hyderabadi-style biryanis(HYB)**, leaving a potential niche unserved. However, the **market analysis revealed a contrasting reality** with approximately **46% of restaurants listed on Zomato** in Othakadai, Madurai also serve **Basmati Biryani**, indicating that the segment is far more competitive than initially assumed. After applying the filtering criteria outlined in Section 2**,** it was identified that **41 restaurants qualify as direct competitors**, effectively challenging the initial positioning premise.

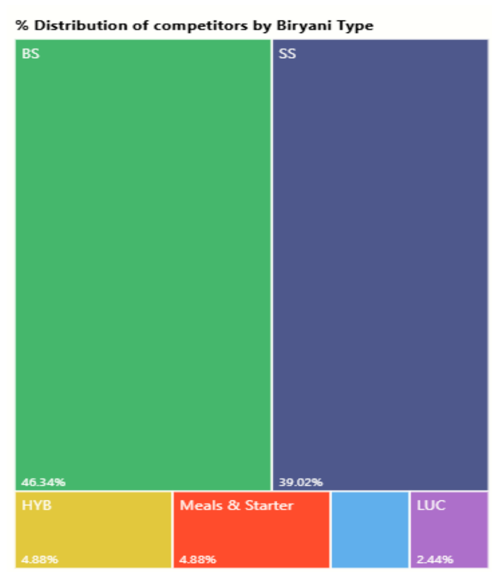



*Fig 18 Distribution of competitors by Biryani Type*

*A. Price-Based Competition*

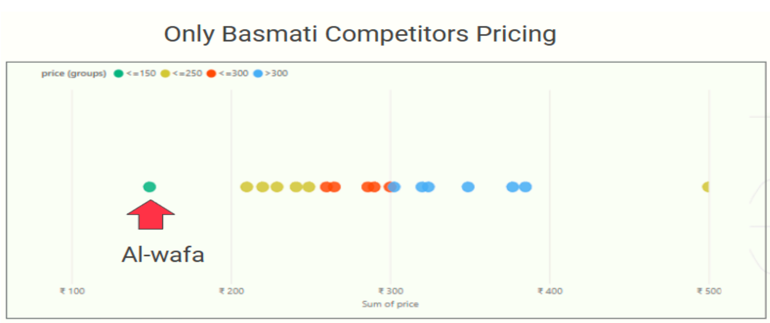



*Fig 19 Clusters by Competitor's Biryani Selling Price*

Price analysis showed that **most competitors price their biryani above ₹200**, placing Ax-Wxxx (average price below ₹200) in a relatively underrepresented and **price-sensitive customer segment** (Refer Fig 19).

This positioning offers a **competitive edge** to attract value-driven customers; however, sustaining this advantage will depend on maintaining consistent quality and perceived value. Notably, within the Basmati Biryani category, **Ax-Wxxx is the only restaurant priced below ₹200**, emphasizing its affordability. Yet, affordability alone may not be sufficient as **price perception must align with customer satisfaction** to prevent the brand from being viewed as “budget” rather than “value for money.”

*B. Rating and Reputation Analysis*

Across the competitor landscape, no restaurant recorded an average rating below 3.0, and **most high-performing competitors maintain ratings above 3.8** (Fig 20) with Ax-Wxxx's current rating of **3.6** indicating moderate customer satisfaction but also signals an opportunity for improvement.

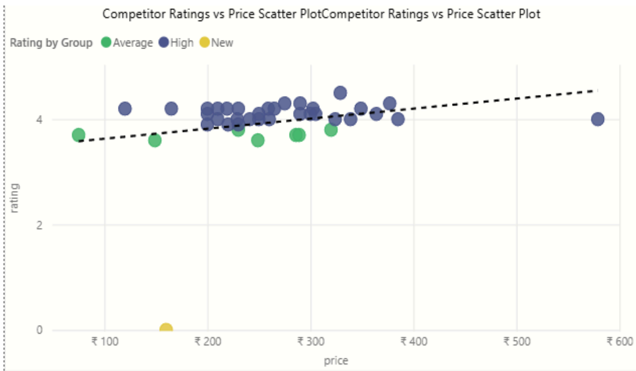



*Fig 20 Competitor's Rating Vs Biryani Selling Price Scatter plot*

An important observation is that **most restaurants tend to maintain more stable average ratings  due to higher numbers of reviews**, as broader feedback smooths out occasional negative experiences. This suggests that **encouraging more reviews through engagement campaigns** could help improve rating stability while collecting valuable customer insights.

*C. Geographic Analysis*

Spatial mapping of competitor locations revealed several concentrated clusters approximately 5 km away from Ax-Wxxx, indicating that most delivery drivers are drawn toward those high-density zones. Given that the majority of Ax-Wxxx's customers reside within a 2–5 km radius, ensuring sufficient driver presence in this local zone becomes operationally crucial for reducing order delays and maintaining delivery reliability. To address this, Ax-Wxxx can implement localized driver-retention strategies that make the outlet a preferred resting and staging point for drivers. Offering small conveniences can go a long way in building loyalty and ensuring proximity during high-demand periods.

### 3.5 Digital Performance Insights

Customer Funnel Metric-Fig 21 (Aug 2024–Jan 2025) reveals a clear divergence between visibility and conversion performance. While impressions increased steadily from 6,998 in August to 10,118 in January, reflecting growing awareness and reach, the menu-to-order conversion rate declined sharply from 16.2% to 11.3%. This indicates that although more potential customers are discovering the outlet, fewer are completing purchases after viewing the menu.

| Metric | Aug 2024 to Jan 2025 trend | Aug 2024 | Sep 2024 | Oct 2024 | Nov 2024 | Dec 2024 | Jan 2025 |
|---|---|---|---|---|---|---|---|
| Customer funnel | | | | | | | |
| Impressions | | 6,998 | 6,248 | 7,626 | 9,374 | 8,513 | 10,118 |
| Impressions to menu | | 19.2% | 17.9% | 16.2% | 16.4% | 15.2% | 15.5% |
| Menu to order | | 16.2% | 19.4% | 19.2% | 16.9% | 13.6% | 11.3% |

*Fig 21 – Customer Funnel Metric from Zomato Dashboard*

This trend highlights a critical gap in digital storefront optimization particularly in how the brand's offerings are **presented and perceived** on delivery platforms. Moreover, the absence of **"Chennai Biryani"** in any menu item reduces the brand's distinctiveness, making it appear generic to users.

## 4. Interpretation of Results and Recommendations

### 4.1 Interpretation of Results

The comprehensive analysis reveals that Ax-Wxxx Bxxxxxx's core challenge lies in impressions-to-order rate, customer retention and brand differentiation within a highly competitive market. While the business successfully positioned itself as an affordable Basmati Biryani provider, the assumption of limited competition was disproven with 46% of restaurants on Zomato serving Basmati Biryani and 41 restaurants as direct competitors.

Despite a strong value proposition under ₹200, Ax-Wxxx faces challenges in sustaining customer loyalty, reflected in a 96.43% churn rate and an average Customer Lifetime Value (CLV) of ₹317.20. Moreover, Zomato rating stability (3.6) and geographic operational constraints, such as limited driver availability in the delivery radius, also hinder growth.

Collectively, these *findings* highlight the need for digital storefront optimization, customer experience enhancement, and local ecosystem strengthening to improve retention, reputation, and operational efficiency.

### 4.2 Recommendations:

*Urgent (0–3 Months): Strengthen Brand Identity, Improve Perceived Value, and Optimize Digital Performance*

Over the next three months, Ax-Wxxx should prioritize building a strong **Chennai-style brand identity**, enhancing **taste and portion consistency**, and improving its **digital storefront performance**. These initiatives are low-cost yet high-impact, aligning closely with customer expectations and online visibility metrics. To achieve measurable improvements, the following **actions** are recommended:

- **Standardize taste and portion sizes** by implementing fixed recipes and consistent packaging containers across all biryani and combo variants to ensure reliable quality and customer satisfaction.
- **Highlight best-performing dishes** such as Chicken Biryani with "Top Rated" or "Speciality" tags to build trust and guide customer choice.
- **Upgrade digital presentation** on Zomato by using professional, consistent food photos showing full portions including all side dishes like Brinjal curry to avoid the perception of low value and quantity.
- **Clarify portion sizes and servings:**

    - Clearly indicate both **weight (e.g., 250g, 500g, 1kg)** & **serving size (e.g., "Serves 1," "Serves 2–3")** in the item descriptions.
    - For side dishes like **Chicken 65**, specify the **number of pieces** and ensure images match the stated portion.
    - Differentiate variants (e.g., *Chicken 65 [120g]*, *Chicken 65 [250g]*, *Chicken 65 [500g]*) with unique images and detailed descriptions. Avoid using identical images for different portion sizes, as it misleads customers to choose the lower-priced option, potentially leading to disappointment and lower ratings.
- **Incorporate Chennai identity** in menu item names and descriptions using local slang (e.g., "*Singaara Samayal Chennai Biryani*", *"Top aroma vera level! Long, soft 'Singaara Samayal' rice slow-cooked with secret masalas that make you say 'Semma tasty da!'"*) and ensure "Chennai Biryani" is explicitly mentioned for regional recall.
- **Improve Order Accuracy** by Conducting **quarterly refresher training** for order-handling and packing staff to ensure accurate communication between order handler and packing teams.
- **Implement a 3rd party order cross check before handover:** The proposed order workflow introduces a structured verification layer to minimize errors and reduce missing items, maintaining Kitchen Preparation Time within the 6–10 minute range. The Order Taker continues to record incoming orders and coordinate with the Packer for preparation and packaging. After packing, the Packer hands the order back to the Order Taker for a quick cross-check against the order note. Once verified, the Order Taker hands over the confirmed package to the Delivery Delivery Partner.

Collectively, these efforts aim to achieve a **10–15% improvement in menu click-to-order conversions** and at least a **0.3-point increase in average ratings** within three months. The plan is achievable using existing kitchen processes and merchant dashboard tools, requiring no major financial investment. By aligning taste reliability, visual consistency, and regional storytelling, Ax-Wxxx can strengthen its emotional connection with customers, enhance online visibility, and convert more first-time buyers into repeat loyal customers.

*Long-Term (6–12 Months):*

**1. Strengthening brand identity & Rating encouragement**

- Launch Chennai-themed packaging that will reinforce recall and authenticity.
- Include QR codes for instant rating collection or social media links.
- Add printed or handwritten rating encouragement notes to increase post-order engagement.

This could increase review volume by **30%** and raise average rating by 0.4 points within 12 months and further strengthen relationships with one-time buyers improving retention.

**2. Data-Driven Menu Innovation & Seasonal Optimization**

Data insights show a strong attachment to 'Chicken 65' based items (Page 12), presenting an opportunity for menu innovation to:

- Develop and test a "Double Chicken 65 Special" combo meal based on strong customer preference data by Month 8,
- Track seasonal consumption trends (e.g., reduced soft drink demand in winter) and adjust the menu accordingly such as introducing soups or warm beverages during colder months.

Adopting limited-period items based on seasonal demand can lift add-on sales by 20% and overall order value by 15%, while refreshing the menu experience for repeat customers.

**3. Weekday Promotions Targeting Workplace & Student Segments**

Weekday performance can be further stabilized through targeted promotions for nearby **workplace and institutional clusters**, such as the Madurai Court and surrounding colleges.

- Promote Pre-Lunch booking options to avoid order delays in peak hours.
- Introduce **Quick-Meal/1+1 offers** targeting the professionals near the Madurai Court and surrounding offices based on the insight that the most frequently used discount was '*Buy 1 Get 1 @ ₹149'* with 291 orders, which likely encouraged bulk purchases in the past.
- Offer **Friday Group Bookings** to promote Combos & bucket orders.
- Launch **result-based student discounts** (e.g., "Score 80%? Get 20% Off!") to build emotional engagement and encourage call-to-order to reduce aggregator dependencies.

The communication of these offers is easily achievable, as the business owner already maintains active WhatsApp groups with nearby colleges and professional communities. Also these efforts could drive a 20–25% increase in weekday orders and 15% growth in repeat purchases, balancing overall weekly sales distribution.

**4. Tackling Delivery Partner Unavailability Affecting Online Status**

To tackle the issue of delivery partner unavailability, a proactive approach is proposed, given the current lack of sufficient data on its root causes. Drawing insights from *"Placing Platform Workers in the City's Urban Imagination"*, it is clear that delivery partners face systemic challenges affecting their presence and reliability.

Based on owners' past experience with the aggregator complaint management system, instead of relying solely on aggregators to resolve these issues Ax-Wxxx can take small, practical steps to build a more supportive local ecosystem. Establishing a modest comfort zone with shaded seating and mobile charging points can encourage delivery partners to stay nearby during peak hours, improving pickup efficiency and reducing delays. Although quantitative evidence is limited, anecdotal feedback such as repeated calls from nearby college customers reporting temporary unavailability suggests that such simple, on-ground initiatives could meaningfully enhance operational reliability and foster a mutually beneficial relationship with delivery partners.

**8. All document on Analysis and references is available here:**